

ABBA4 Implicit Jacobian Formulas

Local implicit-function factors and the triple-jump product

Reduced physical branch for non-autonomous GC2D dynamics

1 Composition notation

Let

$$\gamma = \frac{1}{2 - 2^{1/3}}, \quad \delta = -\frac{2^{1/3}}{2 - 2^{1/3}}, \quad (c_1, c_2, c_3) = (\gamma, \delta, \gamma).$$

For outer step size h , define

$$\ell_j = c_j h, \quad \tau_j = t_n + h \sum_{r < j} c_r, \quad s_j = \frac{\ell_j}{2}, \quad j = 1, 2, 3.$$

The signed physical substeps are

$$z^{[0]} = z_n, \quad z^{[j]} = \Psi_{\ell_j, \tau_j}^{[2]}(z^{[j-1]}), \quad z_{n+1} = z^{[3]}.$$

Every quantity below with first index j belongs to one independent projection solve. In particular, the three multipliers μ_j and their stage points must not be shared.

2 Stage Jacobians for one signed substep

Let

$$W_{j,i} = D_z f(\text{stage point } i, \text{stage time } i), \quad i = 1, 2, 3, 4,$$

and define

$$S_j = W_{j,2} + W_{j,3}.$$

The matrix-product order must be preserved. The exact duplicated ABBA Jacobian has 2×2 blocks

$$J_{\text{ABBA},j} = \begin{pmatrix} A_j & B_j \\ C_j & D_j \end{pmatrix},$$

where

$$\begin{aligned} A_j &= I_2 + s_j^2 W_{j,4} S_j, \\ B_j &= s_j (W_{j,1} + W_{j,4}) + s_j^3 W_{j,4} S_j W_{j,1}, \\ C_j &= s_j S_j, \\ D_j &= I_2 + s_j^2 S_j W_{j,1}. \end{aligned}$$

These expressions are valid without modification for the negative middle substep because $s_2 = \delta h/2 < 0$.

3 Residual derivatives K_j and L_j

For the reduced projection residual

$$R_j(z, \mu) = G \Phi_{\ell_j, \tau_j}^{\text{ABBA}}(Ez + G^T \mu) + 2\mu,$$

define

$$K_j = D_\mu R_j, \quad L_j = D_z R_j.$$

Their compact block expressions are

$$K_j = A_j - B_j - C_j + D_j + 2I_2,$$

$$L_j = A_j + B_j - C_j - D_j.$$

Equivalently,

$$K_j = 4I_2 - s_j(W_{j,1} + W_{j,2} + W_{j,3} + W_{j,4}) \\ + s_j^2(W_{j,4}S_j + S_jW_{j,1}) - s_j^3W_{j,4}S_jW_{j,1},$$

and

$$L_j = s_j(W_{j,1} - W_{j,2} - W_{j,3} + W_{j,4}) \\ + s_j^2(W_{j,4}S_j - S_jW_{j,1}) + s_j^3W_{j,4}S_jW_{j,1}.$$

The converged multiplier is a function

$$\mu_j^* = \mu_{\ell_j, \tau_j}(z^{[j-1]}).$$

Differentiating $R_j(z, \mu_j^*(z)) = 0$ gives

$$L_j + K_j M_j = 0, \quad M_j := D_z \mu_j^*,$$

and hence

$$M_j = -K_j^{-1} L_j.$$

An implementation solves $K_j M_j = -L_j$ and does not form the inverse.

4 Physical Jacobian of one substep

Define

$$P_j = A_j + B_j, \quad Q_j = A_j - B_j + I_2.$$

In expanded form,

$$P_j = I_2 + s_j(W_{j,1} + W_{j,4}) + s_j^2W_{j,4}S_j + s_j^3W_{j,4}S_jW_{j,1},$$

$$Q_j = 2I_2 - s_j(W_{j,1} + W_{j,4}) + s_j^2W_{j,4}S_j - s_j^3W_{j,4}S_jW_{j,1}.$$

The exact ideal-root physical Jacobian is

$$J_j := D_z \Psi_{\ell_j, \tau_j}^{[2]}(z^{[j-1]}) = P_j - Q_j K_j^{-1} L_j.$$

To construct it stably, solve the two-right-hand-side system

$$K_j T_j = L_j, \quad J_j = P_j - Q_j T_j.$$

The Newton matrix K_j is not the physical Jacobian J_j .

5 Equivalent stage-increment factorization

For one substep, the physical mean is

$$z^{[j]} = z^{[j-1]} + \frac{\ell_j}{4}(k_{j,1} + k_{j,2} + k_{j,3} + k_{j,4}).$$

The total derivatives, including $M_j = D\mu_j^*$, obey

$$\begin{aligned} Dk_{j,1} &= W_{j,1}(I_2 - M_j), \\ Du_j^{(1)} &= I_2 + M_j + s_j Dk_{j,1}, \\ Dk_{j,2} &= W_{j,2} Du_j^{(1)}, \\ Dk_{j,3} &= W_{j,3} Du_j^{(1)}, \\ Dv_j^{(2)} &= I_2 - M_j + s_j(Dk_{j,2} + Dk_{j,3}), \\ Dk_{j,4} &= W_{j,4} Dv_j^{(2)}. \end{aligned}$$

Therefore,

$$J_j = I_2 + \frac{\ell_j}{4} \sum_{i=1}^4 Dk_{j,i}.$$

This stage-increment recurrence and the implicit-function formula are two factorizations of the same local tangent. Omitting M_j differentiates a different, explicitly averaged map.

6 Complete ABBA4 Jacobian

The outer chain rule gives

$$D_z \Psi_{h,t_n}^{[4]}(z_n) = J_3 J_2 J_1.$$

The chronological order matters: J_1 acts first and J_3 multiplies from the left. No derivative of the Newton or Broyden iteration history appears; the formulas differentiate the converged implicit roots.

For a tangent vector ξ_n , apply each local Jacobian without constructing the full outer matrix. Set $\eta_0 = \xi_n$ and, for $j = 1, 2, 3$, compute

$$b_j = L_j \eta_{j-1}, \quad K_j y_j = b_j, \quad \eta_j = P_j \eta_{j-1} - Q_j y_j.$$

Then

$$\xi_{n+1} = \eta_3.$$

Each application adds one 2×2 linear solve per particle and no new nonlinear solve.

7 Accumulation over many outer steps

Let $\mathcal{J}_n$ be the local ABBA4 Jacobian for outer step n , and let $\mathcal{M}_n$ be the derivative of the numerical state with respect to the initial condition. Initialize

$$\mathcal{M}_0 = I_2$$

and update

$$\mathcal{M}_{n+1} = \mathcal{J}_n \mathcal{M}_n, \quad \mathcal{J}_n = J_{n,3} J_{n,2} J_{n,1}.$$

Thus the most recent local Jacobian always multiplies from the left.

For the ideal projected roots,

$$J_{n,j}^T J_0 J_{n,j} = J_0$$

for every signed substep, and consequently

$$\mathcal{J}_n^T J_0 \mathcal{J}_n = J_0, \quad \mathcal{M}_n^T J_0 \mathcal{M}_n = J_0.$$

In two dimensions these identities are equivalent to unit determinant. A finite nonlinear tolerance makes the emitted numerical map differ slightly from the ideal-root map evaluated by these formulas.

8 Implementation checklist

For each accepted outer step:

- retain the ordered three substep snapshots and their independent multipliers;
- evaluate $W_{j,1}, \dots, W_{j,4}$ at the signed stage times;
- form K_j, L_j, P_j, Q_j without commuting matrix products;
- solve $K_j T_j = L_j$ or apply the tangent through one right-hand side;
- compose in the order $J_3 J_2 J_1$;
- compare against centered finite differences only at a scale larger than the three nonlinear residual floors.